Flow Matching

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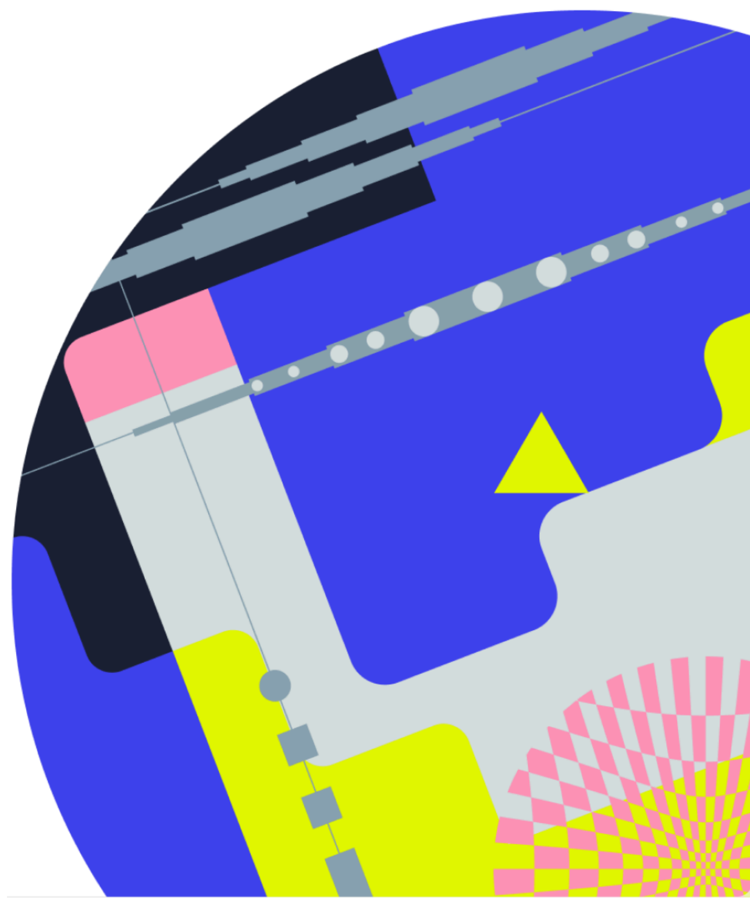


Evolution of density (recap)

- $dX_t = f(X_t, t)dt$

$$\boxed{\frac{\partial}{\partial t} p_t(x) = -\frac{\partial}{\partial x}\left(p_t(x) f(x,t)\right)}$$

Continuity equation

- $dX_t = f(X_t, t)dt + g(t)dW_t$

$$\boxed{\frac{\partial}{\partial t} p_t(x) = -\frac{\partial}{\partial x}\left(p_t(x) f(x,t)\right) + \frac{g^2(t)}{2}\frac{\partial^2}{\partial x^2} p_t(x)}$$

Fokker-Planck(-Kolmogorov) equation

Forward and backward (recap)

$$dX_t = f(X_t, t)\,dt + g(t)\,dW_t$$

$$\downarrow$$

$$dY_t = \left(f(Y_t, t) - \frac{g^2(t)}{2} \nabla \log p_t(Y_t)\right) dt$$

$$\downarrow$$

$$dY_t^{(b)} = \left(-f(Y_t^{(b)}, 1-t) + \frac{g^2(1-t)}{2} \nabla \log p_{1-t}(Y_t^{(b)})\right) dt$$

$$\downarrow$$

$$dX_t^{(b)} = \left(-f(X_t^{(b)}, 1-t) + g^2(1-t) \nabla \log p_{1-t}(X_t^{(b)})\right) dt + g(1-t)\,dW_t$$

$$p_{X_t} = p_{Y_t} = p_{Y^{(b)}_{1-t}} = p_{X^{(b)}_{1-t}} \implies p_{Y_1^{(b)}} = p_{X_1^{(b)}} = p_{data}$$

VP/VE-SDE (recap)

$$dX_t = -\frac{\beta_t}{2} X_t \, dt + \sqrt{\beta_t} \, dW_t \Rightarrow X_t = \sqrt{\bar{\alpha}_t} X_0 + \sqrt{1-\bar{\alpha}_t} \, \varepsilon$$

$$dX_t = g(t) \, dW_t \Rightarrow X_t = X_0 + \sigma_t \varepsilon$$

General case

For each α_t, σ_t $\exists$ $f(t), g(t)$ such that

$$dX_t = f(t) X_t \, dt + g(t) \, dW_t \rightarrow X_t = \alpha_t X_0 + \sigma_t \varepsilon .$$

PF-ODE (recap)

$$dX_t = f(t)X_t\,dt + g(t)\,dW_t$$

$$dY_t = \left(f(t)Y_t - \frac{g^2(t)}{2}\nabla \log p_t(Y_t)\right)dt$$

Sampling: solve from $t=1$ to $t=0$.

Denoising Score Matching (recap)

Loss: $X_0 \sim p_{data}$, $\varepsilon \sim \mathcal{N}(0, I)$, $X_t = \alpha_t X_0 + \sigma_t \varepsilon$

$$\int_0^1 \mathbb{E} \left\| s_\theta(X_t, t) - \frac{\partial}{\partial x} \log p_t(X_t | X_0) \right\|^2 dt \to \min_\theta$$

- $p_{t|0}(x_t | x_0) = \mathcal{N}(x_t | \alpha_t x_0, \sigma_t^2 I) = C \cdot \exp\left(-\frac{1}{2\sigma_t^2} \|x_t - \alpha_t x_0\|^2\right)$

$$\frac{\partial}{\partial x_t} \log p_{t|0}(x_t | x_0) = -\frac{1}{\sigma_t^2}(x_t - \alpha_t x_0) = -\frac{\varepsilon}{\sigma_t}$$

- Parameterize $s_\theta(x_t, t) = \dfrac{-\varepsilon_\theta(x_t, t)}{\sigma_t}$

Loss: $\displaystyle\int_0^1 \omega_t \mathbb{E} \| \varepsilon_\theta(X_t, t) - \varepsilon \|^2 dt \to \min_\theta$

Diffusion vs Flow Matching derivation

	Diffusion	Flow Matching
Forward process $dX_t = f(t)X_t\,dt + g(t)\,dW_t$	Source of theory	
PF-ODE $dY_t = \left(f(t)Y_t - \frac{g^2(t)}{2}\frac{\partial}{\partial x}\log p_t(Y_t)\right)dt$	Learned process	Learned process
Interpolation $X_t = \alpha_t X_0 + \sigma_t \varepsilon$	Convenient sampling	Source of theory

Diffusion vs Flow Matching derivation

	Diffusion	Flow Matching
Known	Fixed $x_0 \rightarrow$ easy $\nabla \log p_{t\|0}(x_t\|x_0)$	Fixed $x_0, \varepsilon \rightarrow$ easy $x_t = \alpha_t x_0 + \sigma_t \varepsilon$ $\Updownarrow$ $dx_t = v_t(x_t\|x_0, \varepsilon)dt$ $v_t(x_t\|x_0, \varepsilon) = \alpha'_t x_0 + \sigma'_t \varepsilon$
Goal	Learn $\nabla \log p_t(x_t)$, sample from PF-ODE $dY_t = \left(f(t)Y_t - \frac{g^2(t)}{2} \nabla \log p_t(Y_t)\right)dt$ that satisfies $p_{Y_t} = p_{X_t}$	Learn $v_t(x)$ s.t. $dY_t = v_t(Y_t)dt$ satisfies $p_{Y_t} = p_{X_t}$

Flow Matching: setting

We know: $dX_t = v_t(X_t | X_0, \varepsilon)dt = (\alpha_t' X_0 + \sigma_t' \varepsilon)dt$

produces dynamics $p_{t|0,\varepsilon}(x_t | x_0, \varepsilon)$.

Goal: find $v_t(x_t)$ s.t. $dY_t = v_t(Y_t)dt$

produces dynamics $p_t(x_t)$.

Key observation:

$$\boxed{\begin{array}{l} dY_t = v_t(Y_t)dt \text{ produces } p_t \iff \\ p_t \text{ and } v_t \text{ satisfy the continuity equation} \\ \frac{\partial}{\partial t} p_t(x) = -\frac{\partial}{\partial x}\left(p_t(x) v_t(x)\right) \end{array}}$$

Flow Matching: derivation

We know: $v_t(x_t|x_0,\varepsilon) = \alpha_t' x_0 + \sigma_t' \varepsilon$ satisfies CE with $p_{t|0,\varepsilon}$.

Goal: find $v_t(x_t)$ that satisfies CE with p_t.

$$\frac{\partial}{\partial t} p_t(x) = \frac{\partial}{\partial t} \int p_{t|0,\varepsilon}(x|x_0,\varepsilon)\, p_{0,\varepsilon}(x_0,\varepsilon)\, dx_0 d\varepsilon =$$

$$= \int \frac{\partial}{\partial t} p_{t|0,\varepsilon}(x|x_0,\varepsilon) \cdot p_{0,\varepsilon}(x_0,\varepsilon)\, dx_0 d\varepsilon = \left[p_{t|0,\varepsilon} \text{ sat. CE with } v_t(x|x_0,\varepsilon) \right]$$

$$= \int -\frac{\partial}{\partial x}\Big(p_{t|0,\varepsilon}(x|x_0,\varepsilon)\, v_t(x|x_0,\varepsilon)\Big)\, p_{0,\varepsilon}(x_0,\varepsilon)\, dx_0 d\varepsilon$$

$$= -\frac{\partial}{\partial x} \int v_t(x|x_0,\varepsilon)\, p_{0,\varepsilon,t}(x_0,\varepsilon,t)\, dx_0 d\varepsilon =$$

$$= -\frac{\partial}{\partial x}\left(p_t(x) \underbrace{\int v_t(x|x_0,\varepsilon)\, p_{0,\varepsilon|t}(x_0,\varepsilon|x)\, dx_0 d\varepsilon}_{v_t(x)} \right)$$

Flow Matching : claim

If $v_t(x_t|x_0,\varepsilon) = \alpha_t' x_0 + \sigma_t' \varepsilon$ satisfies CE with $p_{t|0,\varepsilon}$

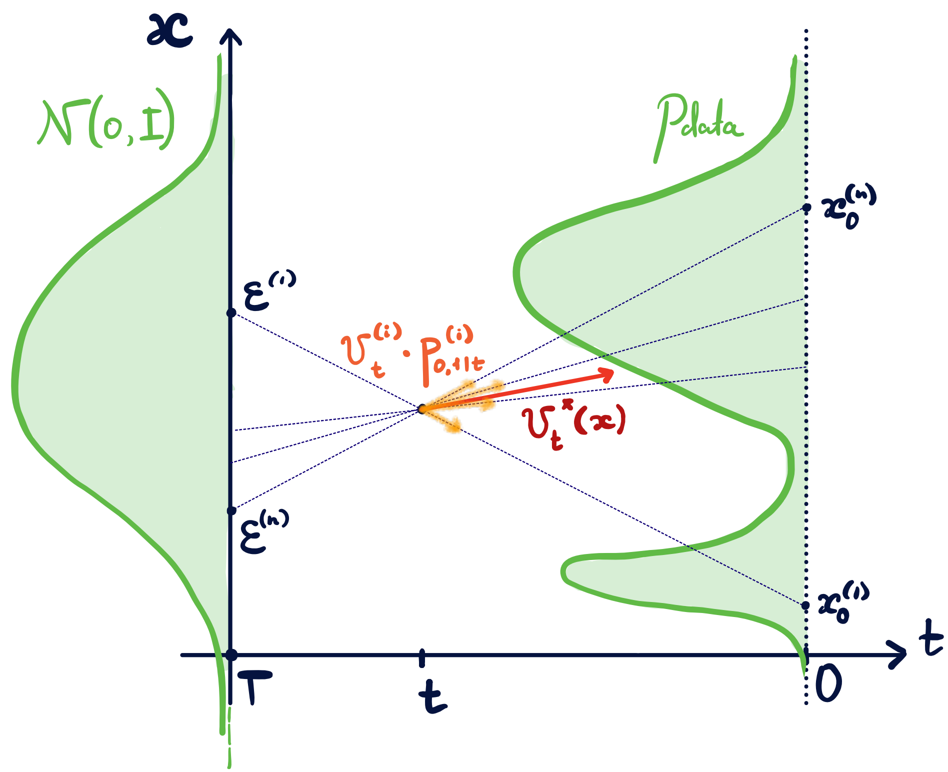
then

$$v_t^*(x) = \int v_t(x|x_0,\varepsilon)\, p_{0,\varepsilon|t}(x_0,\varepsilon|x)\, dx_0 d\varepsilon = \mathbb{E}_{p_{0,\varepsilon|t}(x_0,\varepsilon|x)} v_t(x|x_0,\varepsilon)$$

satisfies CE with p_t.

Interpretation

$v_t^*(x)$: weighted average of directions from ε to x_0 among pairs (x_0,ε) that produce $\alpha_t x_0 + \sigma_t \varepsilon = x$



Flow Matching: training

$$v_t^*(x) = \int v_t(x|x_0,\varepsilon)\, p_{0,\varepsilon|t}(x_0,\varepsilon|x)\, dx_0 d\varepsilon = \mathbb{E}_{p_{0,\varepsilon|t}(x_0,\varepsilon|x)} v_t(x|x_0,\varepsilon)$$

Same old conditional trick

$$\int_0^T \mathbb{E}_{p_t(x_t)} \| v_t^\theta(x_t) - v_t^*(x_t)\|^2 dt = \int_0^T \mathbb{E}_{p_t(x_t)} \|v_t^\theta(x_t)\|^2 dt + \text{const}$$

$$- 2 \int_0^T \mathbb{E}_{p_t(x_t)} \langle v_t^\theta(x_t), v_t^*(x_t)\rangle dt =$$

- $$\mathbb{E}_{p_t(x_t)} \langle v_t^\theta(x_t), v_t^*(x_t)\rangle = \mathbb{E}_{p_t(x_t)} \langle v_t^\theta(x_t), \mathbb{E}_{p_{0,\varepsilon|t}(x_0,\varepsilon|x_t)} v_t(x_t|x_0,\varepsilon)\rangle =$$
$$= \mathbb{E}_{p_{0,\varepsilon,t}(x_0,\varepsilon,x_t)} \langle v_t^\theta(x_t), v_t(x_t|x_0,\varepsilon)\rangle.$$

$$= \int_0^T \mathbb{E}_{p_{0,\varepsilon,t}(x_0,\varepsilon,x_t)} \| v_t^\theta(x_t) - v_t(x_t|x_0,\varepsilon)\|^2 dt + \text{const}$$

Flow Matching vs Score Matching

$$\int_0^T \mathbb{E}_{p_{0,\varepsilon,t}(x_0,\varepsilon,x_t)} \| v_t^\theta(x_t) - v_t(x_t|x_0,\varepsilon)\|^2 dt =$$

$$x_t = \alpha_t x_0 + \sigma_t \varepsilon \quad , \quad v_t(x_t|x_0,\varepsilon) = \alpha_t' x_0 + \sigma_t' \varepsilon .$$

$$= \int_0^T \mathbb{E}_{p_{0,\varepsilon,t}(x_0,\varepsilon,x_t)} \| v_t^\theta(x_t) - (\alpha_t' x_0 + \sigma_t' \varepsilon)\|^2 dt =: \mathcal{L}_{FM}$$

$$x_0 = \frac{x_t - \sigma_t \varepsilon}{\alpha_t} \qquad \varepsilon = \frac{x_t - \alpha_t x_0}{\sigma_t}$$

$$v_t^\theta(x_t) = \alpha_t' \frac{x_t - \sigma_t \varepsilon_t^\theta(x_t)}{\alpha_t} + \sigma_t' \varepsilon_t^\theta(x_t) \Rightarrow \underline{\mathcal{L}_{FM} \equiv \| \varepsilon_t^\theta(x_t) - \varepsilon \|^2}$$

$$v_t^\theta(x_t) = \alpha_t' x_0^\theta(x_t) + \sigma_t' \frac{x_t - \alpha_t x_0^\theta(x_t)}{\sigma_t} \Rightarrow \underline{\mathcal{L}_{FM} \equiv \| x_0^\theta(x_t) - x_0 \|^2}$$

equivalent to denoising
score matching

Flow Matching vs diffusion models

Training: $\int_0^T \mathbb{E}_{p_{0,\varepsilon,t}(x_0,\varepsilon,x_t)} \| v_t^\theta(x_t) - v_t(x_t | x_0, \varepsilon) \|^2 dt$

<=> diffusion

Sampling: $dY_t = v_t^\theta(Y_t)dt$ vs $dY_t = \left(f(t)Y_t - \frac{g^2(t)}{2} s_t^\theta(Y_t)\right)dt$?

Claim Theoretical optimum $v_t^*(x)$ is equal to

$f(t)x - \frac{g^2(t)}{2} \nabla \log p_t(x)$ <=> diffusion PF-ODE

Conclusion Flow Matching $\equiv$ Diffusion

- Only in case $p_0(x) = \mathcal{N}(x | \dots)$ and $x_t = \alpha_t x_0 + \sigma_t \varepsilon$
- Extends to image-to-image
- Different point of view: $\alpha_t x_0 + \sigma_t \varepsilon$ is simpler then SDEs.

Linear interpolant and straight trajectories

- $x_t = t\varepsilon + (1-t)x_0 \Rightarrow v_t(x_t | x_0, \varepsilon) = \varepsilon - x_0$
  straight conditional trajectories (motivation behind FM
  straighter $\Rightarrow$ fewer points)

- Are trajectories of $dY_t = v_t^*(Y_t)dt$ straight?

- Let $p_{data}(x_0) = \mathcal{N}(x_0 | 0, I)$.

  Then $x_t = t\varepsilon + (1-t)x_0 \sim \mathcal{N}(0, (t^2 + (1-t)^2)I)$.

  If Y_t is straight, then $p_{Y_0} = p_{Y_1} = \mathcal{N}(0, I) \Rightarrow p_{Y_t} = \mathcal{N}(0, I)$.

  But $p_{Y_t} = p_{x_t} = \mathcal{N}(0, (t^2 + (1-t)^2)I)$.

Generally, no

- More separated and concentrated modes
  $\Rightarrow$ straighter trajectories.